Name: ____ ____________ **/ 25**

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. **[15 points]** Find the derivative of each function. You do not need to simplify your answers.

   a. $g(t) = \cos(2t) + \sin(3t)$

   b. $y = 2\sec(5x^3)$

   c. $f(\theta) = \sqrt{\cos\left(\frac{\pi}{4}\right) - \csc(\theta) + 10}$

   d. $y = \tan^2(x)\sin^3(x)$

e. $h(x) = \cot^3(x^5 + 2x - 9)$

2. [5 points] Find all x-values where the graph of $f(x) = (x^2 - 6x)^5$ has a horizontal tangent line.

3. [5 points] Find $f''(x)$ if $f(x) = \sin(2x^2)$.